



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

ASSIGNMENT 1

SECI1143

SECTION 03

PROBABILITY & STATISTICAL DATA ANALYSIS

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Assignment 1

Q1

a) Population = entire group of faculty students within the Faculty of Computing.

Sample : a subset of the students population, selected for the study to explore the relationship between faculty students' socioeconomic backgrounds and their academic performance.

b) i) Discrete / continuous

Discrete : Number of courses taken in a semester

Continuous : Grade Point Average (GPA), income

ii) Categorical / Numerical

Categorical : Race , gender

Numerical : Age , GPA , income

iii) Level of measurement :

Nominal : - Gender (male / female)

- Race (Malay / Chinese / Indian / others)

Ordinal : - Academic year (First year, second year)

- Socioeconomic status (low, medium, high)

Interval : - GPA

Ratio : Income

Example of sample data:

Gender: Female

Race: Chinese

GPA: 3.4

Family income: RM 45,000

Academic year: Second year

Socioeconomic status: medium

c) Primary Data Source

- Surveys distributed to students within the Faculty of Computing collecting data directly from the students.

d) Secondary Data Source

- Financial statements, academic performance report (GPA), demographic information and socioeconomic status that contains in the university's record collected during admissions.

e) ~~Possible descriptive statistics~~

~~Mean GPA: 3.3~~

~~Median family income: RM 40,000~~

~~Gender mode: Female~~

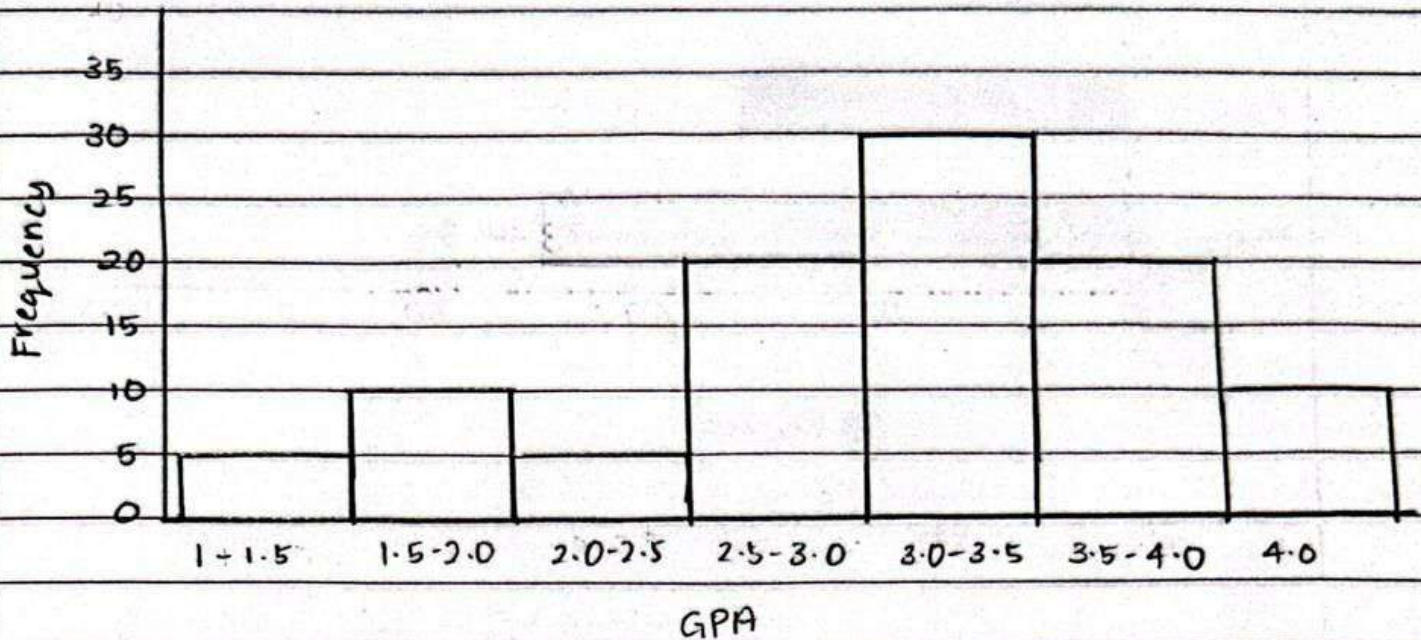
~~Percentage of socioeconomic status: Low (30%)~~

~~medium (60%)~~

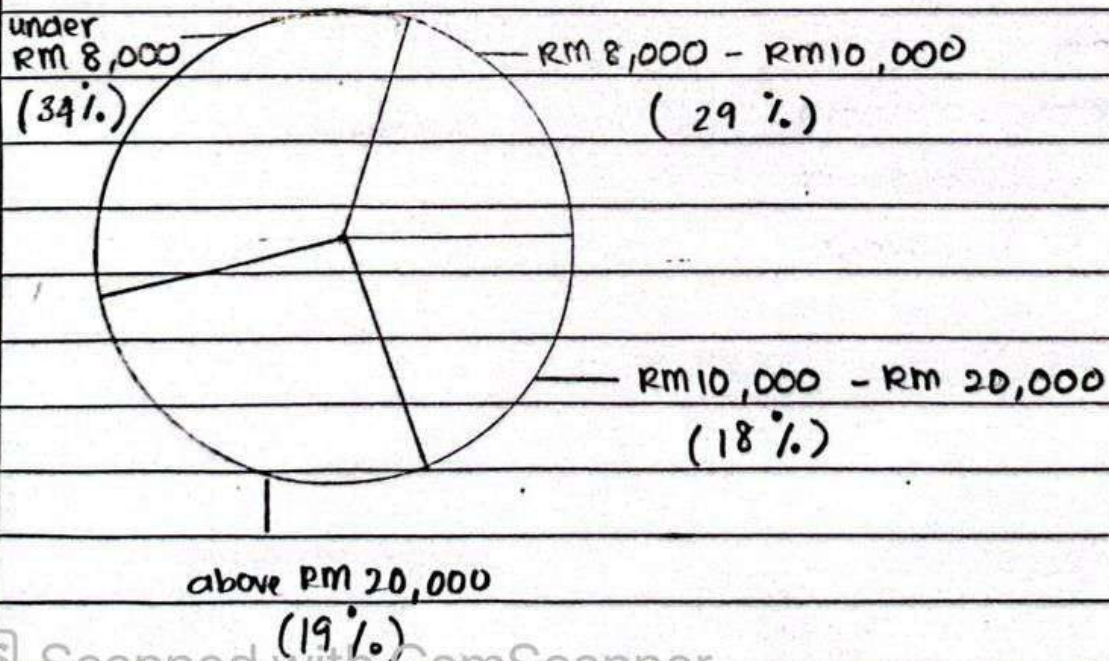
~~High (10%)~~

e) Possible descriptive statistics

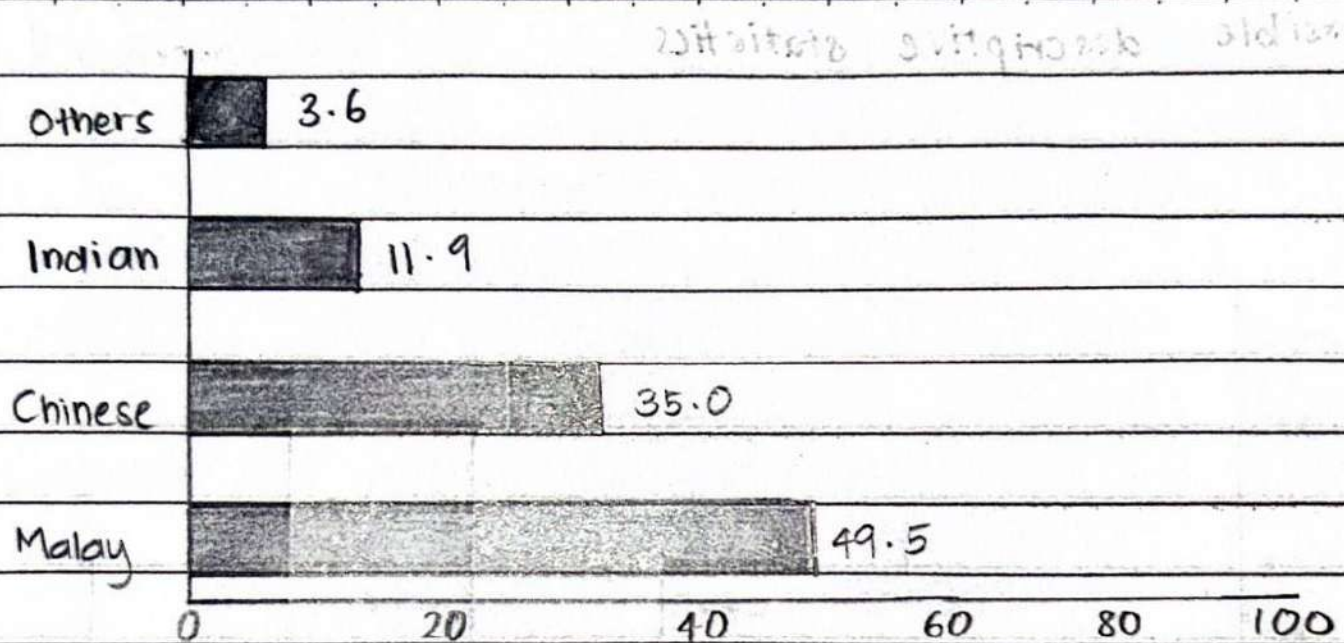
Histogram of GPA for 100 students



Family income of all students (monthly)



Percentage of Race in Faculty of computing, utm



■ Percent

Question 2

3 3 2 0 4 5 6 4 4 3
2 1 2 3 0 5 5 3 2 3
5 4 1 2 0 3 2 4 2 4

$$\text{Relative Frequency} = \frac{\text{frequency}}{\text{total no. of cases}}$$

Number of Students	Tally	Frequency	Relative Frequency
0	III	3	$3 \div 30 = 0.100$
1	II	2	$2 \div 30 = 0.067$
2	IIII II	7	$7 \div 30 = 0.233$
3	IIII II	7	$7 \div 30 = 0.233$
4	IIII	5	$5 \div 30 = 0.167$
5	IIII	4	$4 \div 30 = 0.133$
6	II	2	$2 \div 30 = 0.067$
Total		30	1.00

Question 3

Weight	Tabulation	Frequency
450		10
451		2
452		2
453		7
454		3
455		6
456		4
457		6
458		4
459		3
460		3
461		2
462		2
463		3
464		4
465		3
466		4
467		5
468		0
469		4
470		5

Weight	Tabulation	Frequency
471		8
472		9
473		1

Range

$$X_h = 473$$

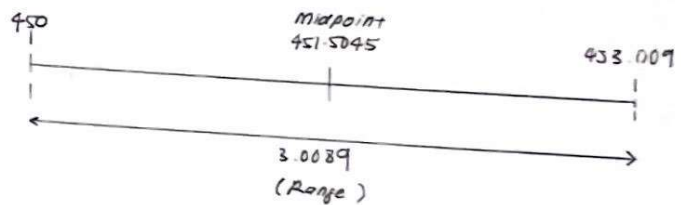
$$X_l = 450$$

$$\begin{aligned} R &= X_h - X_l \\ &= 473 - 450 \\ &= 23 \end{aligned}$$

$$\begin{aligned} \text{Sturges' rule: } i &= \frac{23}{1 + 3.322 \log(100)} \\ i &= 3.0089 \end{aligned}$$

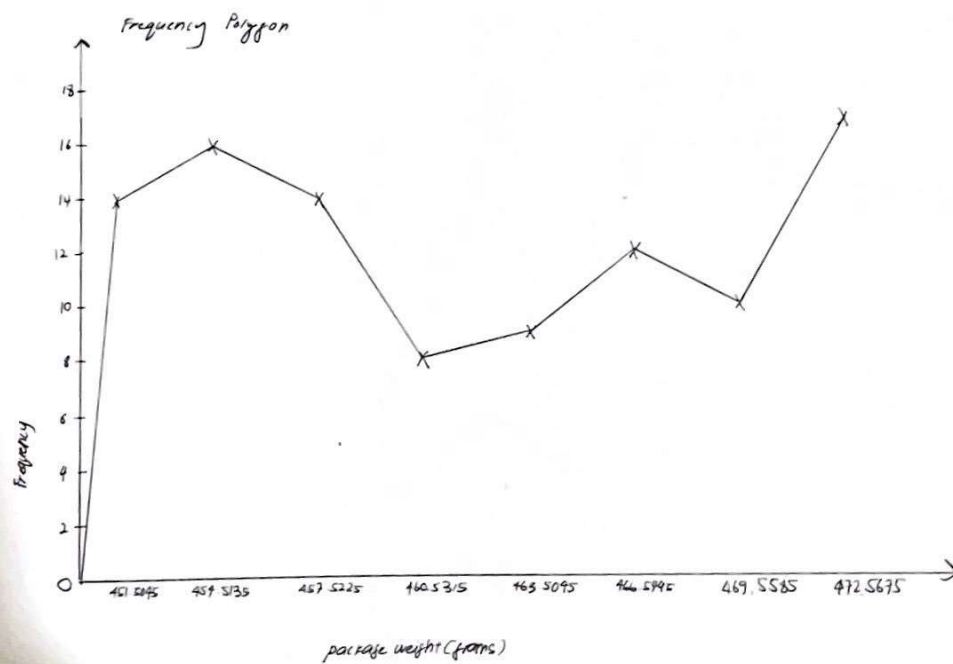
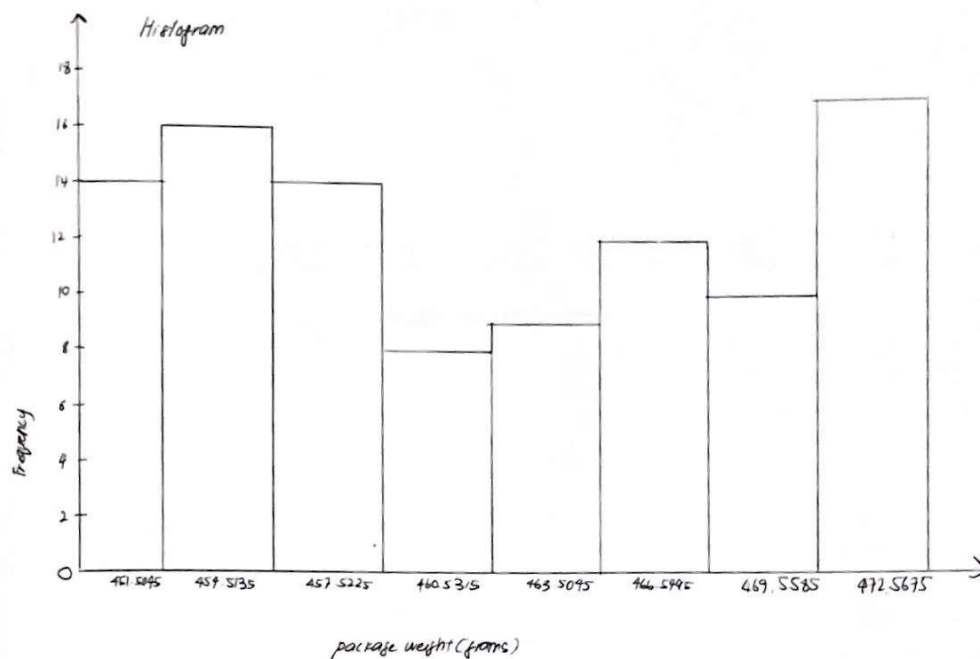
$$\begin{aligned} MP_2 &= X_l + \frac{i}{2} \\ &= 450 + \frac{3.0089}{2} \\ &= 451.5045 \end{aligned}$$

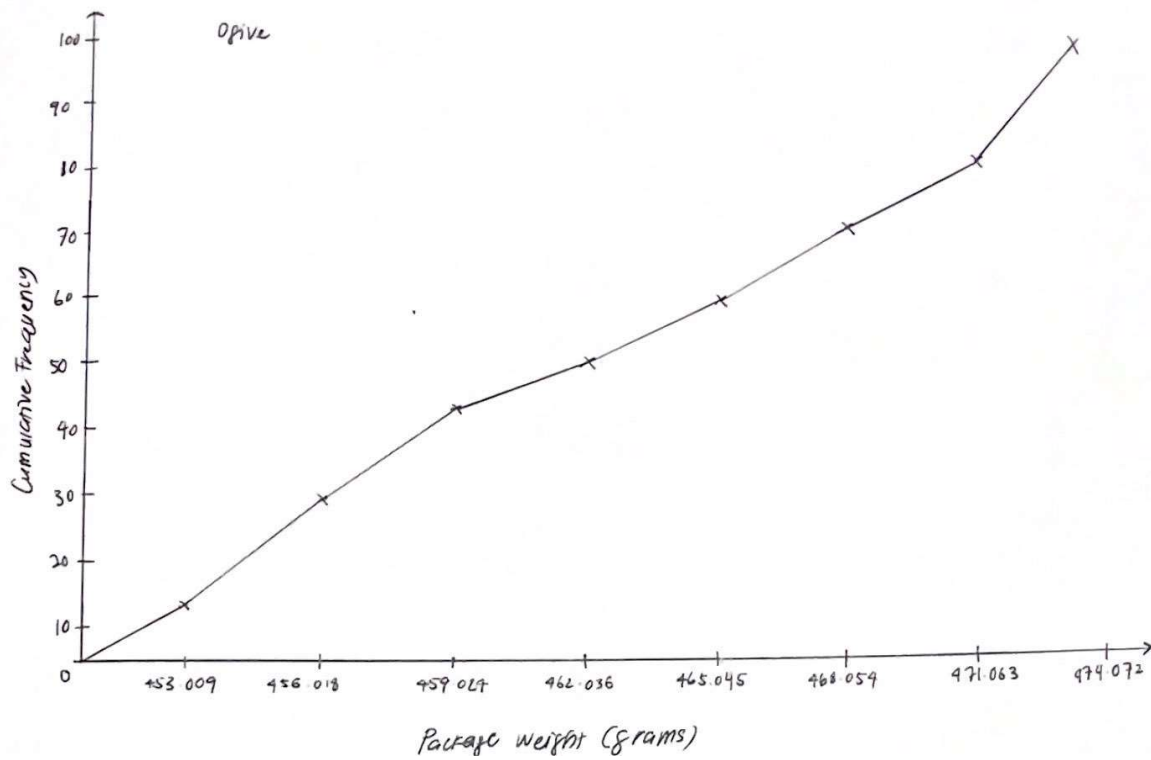
$$\begin{aligned} \text{Midpoint} &= 451.5045 \\ \text{Range} &= 1.0466 \end{aligned}$$



$$\begin{aligned} h &= \frac{23}{3.0089} \\ &= 7.644 \end{aligned}$$

Cell boundaries	Cell midpoint	Frequency	cf
450 - 453 009	451 5045	14	14
453 009 - 456 018	454 5135	16	30
456 018 - 459 027	457 5225	14	44
459 027 - 462 036	460 5315	8	52
462 036 - 465 045	463 5405	9	61
465 045 - 468 054	466 5495	12	73
468 054 - 471 063	469 5585	10	83
471 063 - 474 072	472 5675	17	100





Question 4

78	702	765	811	832	855	896
902	905	918	919	920	923	929
936	938	948	950	956	958	958
970	972	978	1009	1009	1022	1035
1037	1045	1067	1085	1092	1102	1122
1126	1151	1156	1157	1157	1162	1170
1195	1195	1196	1217	1237	1311	1333
1340						

Q1 = ?

$$N = 50, p = 25, i = \frac{25 \times 50}{100} = 12.5, k = 13$$

$$Y[13] = 923 \quad \therefore Q1 = 923$$

Q2 = ?

$$N = 50, p = 50, i = \frac{50 \times 50}{100} = 25, k = 25$$

$$(Y[25] + Y[26]) \div 2 = (1009 + 1009) \div 2 = 1009 \quad \therefore Q2 = 1009$$

Q3 = ?

$$N = 50, p = 75, i = \frac{75 \times 50}{100} = 37.5, k = 38$$

$$Y[38] = 1156 \quad \therefore Q3 = 1156$$

$$\begin{aligned} iqr &= \text{upper quartile} - \text{lower quartile} \\ &= 1156 - 923 \\ &= 233 \end{aligned}$$

$$\begin{aligned} \text{lower limit} &= Q1 - 1.5 \times iqr \\ &= 923 - 1.5 \times 233 \\ &= 573.50 \end{aligned}$$

$$\begin{aligned} \text{upper limit} &= Q3 + 1.5 \times iqr \\ &= 1156 + 1.5 \times 233 \\ &= 1505.50 \end{aligned}$$

⇒ Modified Box Plot

outliers = 78

lower limit = 573.50

$Q_1 = 923$

median = 1009

$Q_3 = 1156$

upper limit = 1505.50

